

Since $p \rightarrow \neg q$ is True, by defn. of conditional, p is False.

Since $r \rightarrow p$ is True, by defn. of conditional, r is False.

p, q, r are all False. All 4 system specifications are consistent.

12. Are these system specifications consistent? "If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer."

Soln: p : File system is ~~not~~ locked q : New messages are queued
 r : System is functioning normally s : New messages will be sent to the message buffer.

① $\neg p \rightarrow q$ ② $\neg p \leftrightarrow r$ ③ $\neg q \rightarrow s$ ④ $\neg p \rightarrow s$ ⑤ $\neg s$ are True.

By ⑤, Defn. of negation, s is False

By ④, Defn. of conditional, $\neg p$ is False $\therefore$ By defn. of conditional, $\neg p \rightarrow q$ is True (consistent with ①)

By ③, Defn. of conditional, $\neg q$ is False

By ②, Defn. of biconditional, r is False

⑥ Defn. of negation, p and q are respectively True.

All 5 system specifications are consistent.

17. Suppose that in Example 7, the inscriptions on Trunks 1, 2 and 3 are "The treasure is in Trunk 3", "The treasure is in Trunk 1", and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.

- "All the inscriptions are false."
- "Exactly one of the inscriptions is true."
- "Exactly two of the inscriptions are true."
- "All three inscriptions are true."